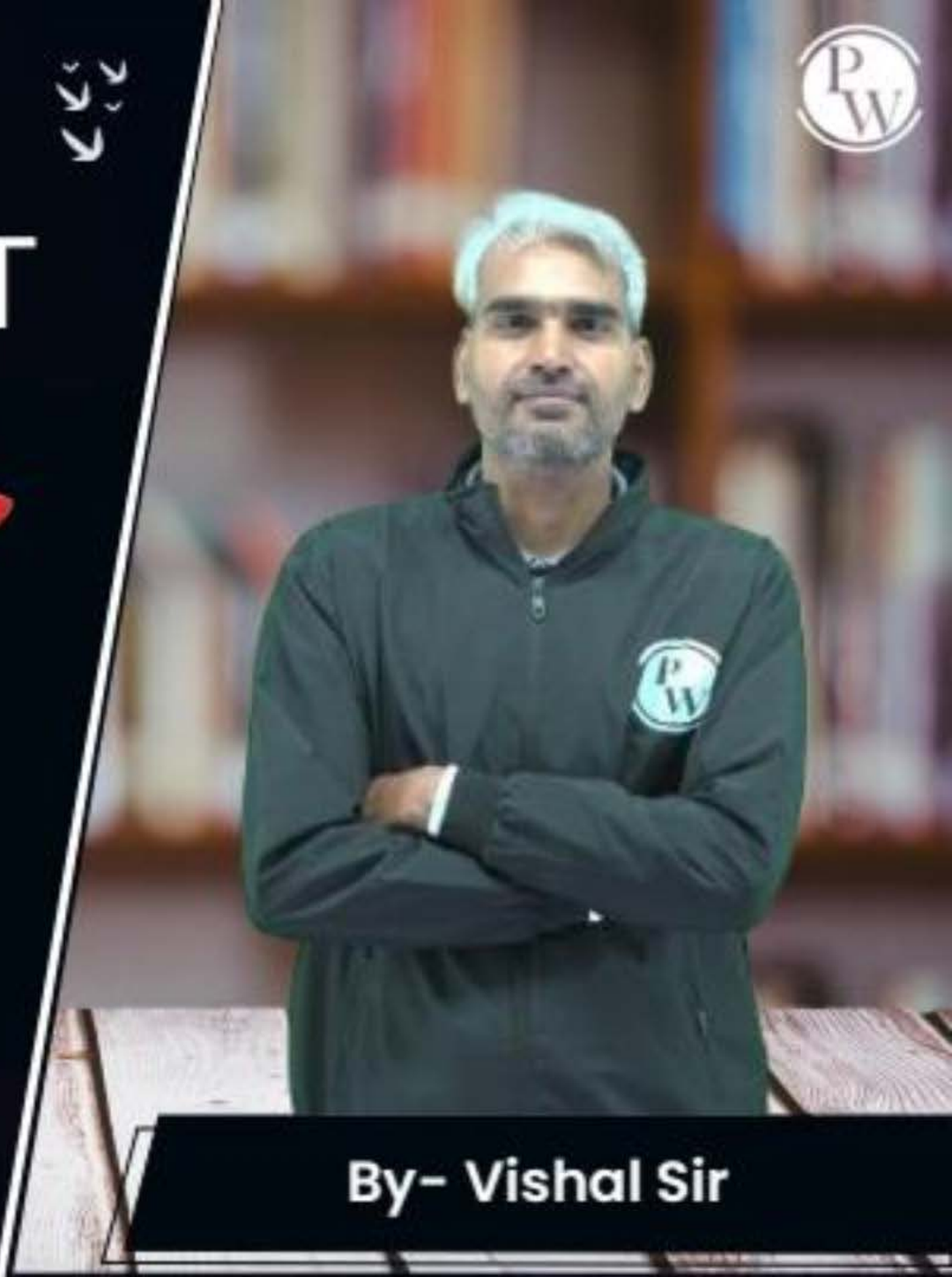
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Functional dependency

Topic Types of functional dependency

Slide

Topics to be Covered



- Topic: Properties of functional dependency
- Topic: Different types of keys in RDBMS
- Topic: Candidate key
- Topic: Super key

Slide

GATE 2024

H.W:

FD $X \rightarrow Y$ is called an useful FD if and only if both X and Y are non-Empty sets and

$$X \cap Y = \phi$$

How many useful FDs are possible in a relation with '4' attributes ?

Q:- How many non-trivial FDs are possible in a relation with "4" attributes

H.W.

R(ABCD)

A → B	C → A	A → BC	C → AB	A → BCD	AB → C, AB → D	AB → CD	ABC → D
A → C	C → B	A → BD	C → AD	B → ACD	AC → B, AC → D	AC → BD	ABD → C
A → D	C → D	A → CD	C → BD	C → ABD	AD → B, AD → C	AD → BC	ACD → B
B → A	D → A	B → AC	D → AB	D → ABC	BC → A, BC → D	BC → AD	BCD → A
B → C	D → B	B → AD	D → AC		BD → A, BD → C	BD → AC	
B → D	D → C	B → CD	D → BC		CD → A, CD → B	CD → AB	
		28			18		4

Total FDs with 4 attributes = 28 + 18 + 4 = 50

Q:- How many non-trivial FDs are possible in a relation with "4" attributes

H.W.

Case ① When in FD $X \rightarrow Y$, Cardinality of X is 1.

$X \longrightarrow Y$

$^4C_1$ and $(^3C_1$ or $^3C_2$ or $^3C_3)$

$4 * (3+3+1) = 4 * 7 = 28$

Case ② When in FD $X \rightarrow Y$, Cardinality of X is 2.

$X \longrightarrow Y$

$(^4C_2)$ and $(^2C_1$ or $^2C_2)$

$6 * (2+1) = 6 * 3 = 18$

Case ③ When in FD $X \rightarrow Y$, Cardinality of X is 3

$X \longrightarrow Y$

$(^4C_3)$ and $(^1C_1)$

$4 * 1 = 4$

Total no. of FDs

$= 28 + 18 + 4$

$= 50$

Q:. Let R(A B C D E) be the relational schema.
How many non-trivial FDs can be defined in relation R

$X \longrightarrow Y$

Case ① $|X| = 1 \Rightarrow {}^5C_1 * ({}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4) = 5 * (4+6+4+1) = 5 * 15 = 75$

Case ② $|X| = 2 \Rightarrow {}^5C_2 * ({}^3C_1 + {}^3C_2 + {}^3C_3) = 10 * (3+3+1) = 10 * 7 = 70$

Case ③ $|X| = 3 \Rightarrow {}^5C_3 * ({}^2C_1 + {}^2C_2) = 10 * (2+1) = 10 * 3 = 30$

Case ④ $|X| = 4 \Rightarrow {}^5C_4 * ({}^1C_1) = 5 * 1 = 5$

∴ Total no. of FDs Possible = 180



Topic : Properties of Functional Dependencies

① Reflexivity :→ If $X \supseteq Y$, then $X \longrightarrow Y$ is called a reflexive FD.

Every Reflexive FD always holds in a relation

② Augmentation :→ Let functional dependency $X \longrightarrow Y$ exists in the relation R, and Z is some set of attributes from relation R, then

$XZ \longrightarrow YZ$ will also exist in rel^n R.

③ Transitivity :→ Let FDs $X \longrightarrow Y$ & $Y \longrightarrow Z$ exists in the relation, R, then FD $X \longrightarrow Z$ will also exist in the relation R.

This three properties are called Armstrong's Axioms



Topic : Properties of Functional Dependencies

④ Decomposition :- (Splitting Rule)

If $X \rightarrow YZ$ exists in relation R, then $X \rightarrow Y$ and $X \rightarrow Z$ both will hold true in relation R.

Note:- If $XY \rightarrow Z$ exists in R,

then $X \rightarrow Z$ & $Y \rightarrow Z$ need not hold true in rel^n R

Let $AB \rightarrow BC$ holds true in rel^n R

Semi non-trivial FD

$AB \rightarrow BC \Rightarrow$ ∴ $AB \rightarrow B$ {trivial Part}

&

$AB \rightarrow C$ {non-trivial}

We can always split a semi non-trivial FD into two Parts, trivial Part & non-trivial Part.



Topic : Properties of Functional Dependencies

⑤ Union :- If $X \rightarrow Y$ and $X \rightarrow Z$ exists in relation R, then $X \rightarrow YZ$ will also hold true in relation R

⑥ Composition :- If $X \rightarrow Y$ and $P \rightarrow Q$ exists in relation R then $XP \rightarrow YQ$ will also hold in relation R

⑦ Pseudo Transitivity :- If $X \rightarrow Y$ and $YW \rightarrow Z$ exists in relation R then $XW \rightarrow Z$ will also hold true in rel^n R

Key Concept :-

In a relational table no two tuples of the table can be exactly same

↳ i.e, in relational table duplicate tuples are not allowed.

↳ To implement this restriction we say that every relation must have a key.

* A key in a relation is the set of attributes that can uniquely identify each tuple of the relation



Topic : Different types of keys

There are various types of keys

1. Candidate key
2. Primary key
3. Alternate key
4. Super key
5. Foreign key

Foreign key Concept will be discussed in the last Capter
i.e: "ER Model & Integrity Constraints"



2 mins Summary

Topic Properties of functional dependency

Topic Different types of keys in RDBMS

Topic Candidate key

Topic Super key

Slide


THANK - YOU
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